Optimization of Water Distribution Network Design Using the Shuffled Frog Leaping Algorithm

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Abstract: *Shuffled Frog Leaping Algorithm* (*SFLA*) is a meta-heuristic for solving discrete optimization problems. Here it is applied to determine optimal discrete pipe sizes for new pipe networks and for network expansions. *SFLA* is a population based, cooperative search metaphor inspired by natural memetics. The algorithm uses memetic evolution in the form of infection of ideas from one individual to another in a local search. The local search is similar in concept to particle swarm optimization. A shuffling strategy allows for the exchange of information between local searches to move toward a global optimum. This paper summarizes the development of *SFLANET*, a computer model that links *SFLA* and the hydraulic simulation software *EPANET* and its library functions. Application of *SFLANET* to literature network design problems is then described. Although the algorithm is in its initial stages of development, promising results were obtained.

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Introduction

Water distribution networks are some of the most expensive infrastructure systems. A water distribution network contains several components: pipes, pumps, and other hydraulic devices, and reservoirs. Each component is designed as a compromise between economic and technical considerations. This paper focuses on the application of a new optimization method to the pipe sizing problem.

Water distribution network design belongs to a group of inherently intractable problems referred to as NP-hard. Essentially NP-hard means that a rigorous algorithm to find an optimum design using discrete diameters is not a practical possibility. It is well known that when diameters are used as the decision variables, the constraints are implicit functions of the decision variables and require solving conservation of mass and energy to determine the network's pressure heads. In addition, the feasible region is nonconvex, and the objective function is multimodal. Hence the network optimization problem remains very challenging.

In recent years, researchers have attempted to exploit expanding computer power and combined new optimization techniques with hydraulic simulation software. This paper describes the development of a computer model named *SFLANET*. *SFLANET* is based upon the *Shuffled Frog Leaping Algorithm* (*SFLA*), a new member in the family of memetic algorithms (a kind of meta-heuristics). The optimization algorithm is linked to *EPANET* (Rossman 2000) via the *EPANET Toolkit* and can be used for pipe network design.

The remainder of the paper is structured as follows. A mathematical formulation of the design problem is presented and techniques that have been applied to solve it are discussed. *SFLA* and *SFLANET* are then introduced and results and comments on the performance of *SFLANET* for some example problems are presented. Two examples are for new network designs and the last, the New York City tunnel problem, has the goal of determining an optimal network expansion. The paper closes with some summary remarks.

Optimization of Water Distribution Networks

Problem Formulation

In this paper, water distribution network design is formulated as a least-cost optimization problem with a selection of pipe sizes as the decision variables. The pipe layout and its connectivity, nodal demand, and minimum head requirements are assumed known. In a general form, the optimization of water distribution networks can be stated as:

Find the least cost combination of pipe sizes while satisfying the following conditions.

- Conservation of mass: Inflows and outflows must balance at each node,
- Conservation of energy: Head loss around a closed loop must equal zero and the head loss along a path between two reservoirs must equal the reservoirs' elevation difference,
- Head loss in each pipe follows a known function of the flow in the pipe, its diameter, length, and hydraulic properties,
- Pressure head and flow requirements: Minimum pressure must be provided at network locations for a given set of demand(s), and
- Acceptable pipe sizes: Diameters are selected from an admissible set.

Mathematically the formulation can be stated as

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$$\min f(D_1,\ldots,D_{npipe}) = \sum_{k \in npipe} C(D_k, L_k) \quad (1)$$

Subject to

$$\sum Q_{\text{in}} - \sum Q_{\text{out}} = Q_e \quad (2)$$

$$\sum_{k \in \text{Loop}l} \Delta H_k = 0, \quad \forall l \in NL \quad (3)$$

$$\Delta H_k = H_{u/s\text{node},k} - H_{d/s\text{node},k} = \omega \frac{L_k}{CW_k^{\beta} D_k^{\gamma}} Q_k |Q_k|^{\beta - 1},$$

$$\forall k \in npipe \quad (4)$$

$$H_{iN} \geqslant H_{\min iN}, \quad \forall iN \in NN \quad (5)$$

$$D_k \in \{D\}, \quad \forall k \in npipe \quad (6)$$

where $D_1,\ldots,D_{npipe} = npipe$ discrete pipe diameter decisions selected from the set of commercial pipe sizes $\{D\}$ [Eq. (6)]; $C_k(D_k, L_k)$ = cost of pipe k with diameter D_k and length L_k. Eq. (2) represents conservation of mass for each node where Q_{in} and Q_{out} = flow into and out of the node, respectively, and Q_e = external inflow (negative) or demand (positive) at the node. Eq. (3) represents the conservation of energy around loop l where ΔH_k denotes the head loss in pipe k and NL = total number of loops in the system. The head loss in each pipe is the head difference between connected nodes [Eq. (4)] where CW = the Hazen–Williams roughness coefficient (the Darcy–Weisbach equation can also be used). Eq. (5) requires that the nodal pressure H for any node iN (where total number of nodes is NN) is equal to or greater than a prespecified minimum pressure $H_{\min}$. For ease of presentation, one demand pattern is shown above. However, this formulation and the solution method can be extended to consider more than one demand patterns and/or extended period simulations.

Previous Studies

The traditional method for designing pipe networks is trial and error guided by experience. Designers make use of commercially available hydraulic simulation packages to solve the network equations [Eqs. (2)–(4)] and rules of thumb are often used to provide direction in the design process. A common technique is to ensure that the slope of the hydraulic grade line or the flow velocity lie within reasonable bounds for each pipe.

Monbaliu et al. (1990) proposed a rule based gradient search technique to achieve an efficient design. Initially, all pipes are set at their minimum diameters and a simulation package is used to determine the pressures at all nodes in the network. If the minimum pressure requirements [Eq. (5)] are not satisfied, the pipe with the maximum head loss per unit length is increased to the next available size and another hydraulic simulation is completed. The process is repeated until all pressure constraints are satisfied. Because of the complex interactions between the system components it is often difficult to ensure the effectiveness of this so-called greedy algorithm.

All possible combinations of available discrete pipe sizes can be simulated in a complete enumeration. One then selects the lowest cost combination that satisfies the design constraints. The main drawback is the amount of time required to complete the task. Since complete enumeration is impractical for even mid-sized systems, Gessler (1985) proposed a selective enumeration method for network design that prunes the search space. He suggested guidelines to eliminate certain inferior solutions from being evaluated by the hydraulic simulation model, hence reducing the number of configurations to be analyzed. However, Murphy and Simpson (1992) showed that the approach may miss the optimal design even for a moderately sized network.

A number of researchers have dealt with the problem of optimization of water distribution networks by applying mathematical programming techniques. Virtually every optimization method has been applied to the water distribution network optimization problem. In many applications, the hydraulic relationships are simplified to a form required for the solution algorithm. These simplifications include linearizing the hydraulic equations or treating the discrete decisions as continuous variables.

Alperovits and Shamir (1977) developed a linearization approach known as the linear programming gradient method. The approach reduces the complexity of the original nonlinear problem by solving a sequence of approximate linear subproblems. The technique requires that a set of variables (pipe flows) be defined and a linear program is formulated to improve the pipe selection for those flow rates. Information available from the solution of the linear programming problem is then used to update the pipe flows. This iterative process continues until an overall convergence criterion is met. This approach has been adapted and improved by many researchers (Quindry et al. 1981; Featherstone and El-Jumaily 1983; Fujiwara et al. 1987).

Although a linear program is solved at each iteration in the methods above, the design problem is nonlinear and global optimality cannot be guaranteed. To overcome this limitation, some researchers (Fujiwara and Khang 1990; Eiger et al. 1994; Loganathan et al. 1995) have developed approaches to move toward global optimal solutions through two-phase searches.

By its nature, the optimization of water distribution networks is nonlinear. Therefore it is reasonable to use standard or application-tailored nonlinear programming methods. A number of researchers have applied nonlinear optimization to pipe network problems (Shamir 1974; El-Bahrawy and Smith 1985; Chiplunkar et al. 1986; El-Bahrawy and Smith 1987; Su et al. 1987; Lansey and Mays 1989). NLP methods have two significant limitations. First, they also do not guarantee global optima. Second, they typically use continuous diameters as decision variables and converting the continuous solutions to commercial pipe sizes may reduce the quality of the solution.

Recently, developments in the field of stochastic optimization have proved to be useful for solving the discrete pipe-sizing problem. Numerous heuristic techniques have been developed. In mathematical programming, heuristic refers to a procedure that seeks a solution to an optimization problem but does not guarantee that it will find one. Heuristics use an objective function (termed as heuristic function) to guide the search, as a human would do. Heuristic searches are of two types; informed search and blind search. Heuristics that use the heuristic function in a strategic way (i.e., an informed search) are referred to as meta-heuristics or stochastic search methods. Genetic algorithms, simulated annealing, and tabu search are common meta-heuristics.

In general, when searching for an optimal solution with a stochastic search method, the objective function is calculated for a set of decision variables (discrete pipe sizes). Based only on the objective function values of the current solutions, the search evolves through generations, improving the features of the potential solutions through some form of evolution. The generation process varies by method but always contains a random element.

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Stochastic search methods only use objective function values to move to better solutions compared to most mathematical programming methods that use gradient information. As such, they can easily be used to solve the discrete pipe-sizing problem. The drawback of these methods is the large number of simulations necessary to find an optimum solution since gradient information about the response surface is not used. Applications to design water distribution networks include genetic algorithms and simulated annealing. Genetic algorithms have been used for solving the network design problem by Hadji and Murphy (1990); Murphy and Simpson (1992); Walters and Cembrowicz (1993); Simpson et al. (1994); Dandy et al. (1996); Savic and Walters (1997); Gupta et al. (1999); Lippai et al. (1999). Simulated annealing was applied by Loganathan et al. (1995) and Cunha and Sousa (1999).

In an effort to reduce the computational burden in solving the problem a new meta-heuristic was applied. The approach, *Shuffled Frog Leaping Algorithm* (*SFLA*), has shown promise in solving other problems (Eusuff and Lansey 2000). The remainder of the paper focuses on *SFLA*, its algorithmic procedure, and its application to literature problems.

Shuffled Frog Leaping Algorithm: Memetic Meta-Heuristic

The *Shuffled Frog Leaping Algorithm* (*SFLA*) is a memetic meta-heuristic that is designed to seek a global optimal solution by performing an informed heuristic search using a heuristic function. It is based on evolution of memes carried by interactive individuals and a global exchange of information among the population. *SFLA* has been tested on several combinatorial problems and found to be efficient in finding global solutions (Eusuff and Lansey 2000).

Basic Concept

Memetic algorithms are population-based approaches for heuristic search in optimization problems. The term memetic comes from ''meme'' (Dawkins 1976). Memes can be considered as the unit of cultural evolution. Ideas evolve in a manner analogous to biological evolution. Here we introduce the term memeplex to denote a group of mutually supporting memes that form an organized belief system, such as a religion.

Evolution: Memetic versus Genetic

Although memetic and genetic evolution are subjected to the same basic principles, i.e., possible solutions are created, selected according to some measure of fitness, combined with other solutions, and possibly mutated, memetic evolution is a much more flexible mechanism. The implicit goals of genes and memes are different to the degree that they use different mechanisms for spreading information from one member of the population to another member. Genes can only be transmitted from parents to offspring. Memes, in principle, can be transmitted between any two individuals. For genes to be transmitted, it typically must occur between generations. Unlike genetic evolution, if an improved idea is found in memetic evolution, it can be incorporated in other memes immediately rather than waiting for a full generation of genes to be replicated. Also, gene replication is restricted by the rather small number of offspring that a single parent can have, whereas the number of individuals that can take over a meme from a single individual is almost unlimited.

Overview of the Shuffled Frog Leaping Algorithm

The *Shuffled Frog Leaping Algorithm* progresses by transforming ''frogs'' in a memetic evolution. In this algorithm, individual frogs are not so important, rather they are seen as hosts for memes and described as a memetic vector. Each meme consists of a number of memotypes. The memotypes represent an idea in a manner similar to a gene representing a trait in a chromosome in a genetic algorithm. The *SFLA* does not change the physical characteristics of an individual rather it progressively improves the ideas held by each frog in a so-called virtual population (VP). The virtual population is used to model the meme pool consisting of a diverse set of frogs in a manner analogous to the population representing a chromosome pool in a genetic algorithm (GA) population. As noted above, the difference and potential advantage of the memetic representation is that ideas are passed between all individuals in the population whereas only parent–sibling interactions are allowed in a GA.

Consider a group of frogs leaping in a swamp. The swamp has a number of stones at discrete locations that the frogs can leap onto. The frogs' goal is to find the stone with the maximum amount of available food as quickly as possible by improving their memes. Here, the alteration of memotypes can be possible only by discrete values and the meme corresponds to the coordinate position of the frogs. The frogs can communicate with each other, and can improve their memes by infecting (passing information) each other. Improvement of memes results in changing an individual frog's position by adjusting its leaping step size.

Based on this abstract model of virtual frogs, the *Shuffled Frog Leaping Algorithm* draws on particle swarm optimization (PSO) (Eberhart and Kennedy 1995) as a local search tool and the idea of competitiveness and mixing information from parallel local searches to move toward a global solution from the *Shuffled Complex Evolution* (*SCE*) algorithm (Duan et al. 1992).

The sample of virtual frogs constitutes a population. The population is partitioned into subsets described as memeplexes. This step is similar to *SCE* but *SFLA* considers discrete variables whereas *SCE* is only applicable to continuous variables. The memeplexes can be perceived as a set of parallel frog cultures attempting to reach some goal. Each frog culture proceeds towards their goal exchanging ideas independently (i.e., search the space in different directions) in parallel. Frog leaping (a virtual memetic engineer) improves an individual's meme and enhances its performance towards the goal. Frog leaping is an extension of the process used in PSO. PSO is developed for continuous problems whereas *FLA* considers discrete variables. Within each memeplex, the frogs are infected by other's ideas, hence they experience a memetic evolution.

After a defined number of memetic evolution steps, information is passed between memeplexes in a shuffling process similar to that used in *SCE*. Shuffling is analogous to the set of cultures that were working in isolation, meeting and transferring information between the cultures. Shuffling enhances the meme quality after being infected by the frogs from different memeplexes. Shuffling ensures that the cultural evolution towards any particu-

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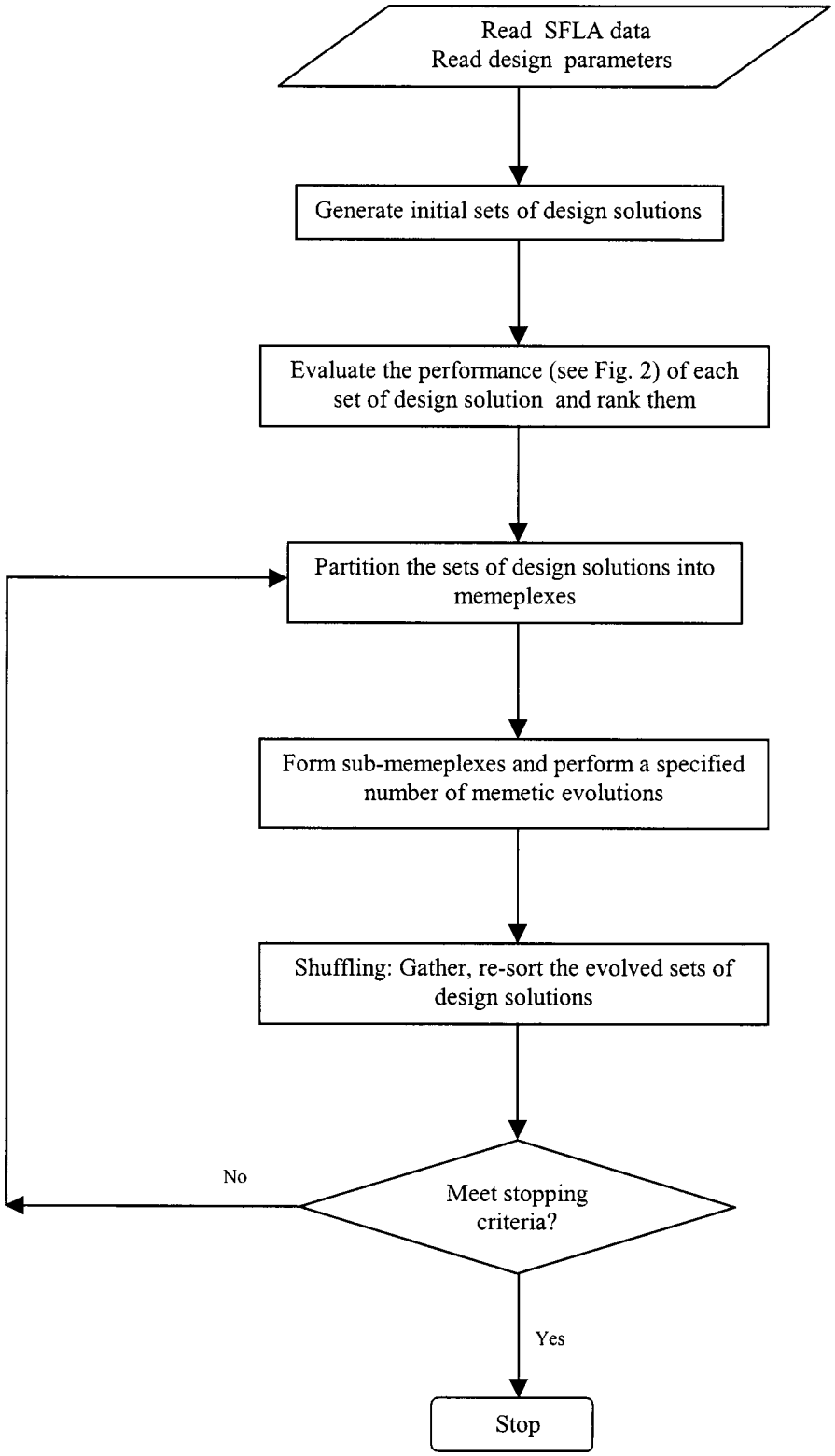


Fig. 1. Solving optimum network design problem with *SFLANET*

lar interest is free from bias. After shuffling, the searches for the optimal solutions using information in the independent memeplexes are again pursued. This local search and shuffling process continues until defined convergence criteria are satisfied. The next section discusses in detail *SFLA* and the network optimization strategy.

Development of *SFLANET*

This section describes the development of *SFLANET*, a computer model that applies *SFLA* in combination with the hydraulic simulation software, *EPANET* (v. 2), and its library functions, *EPANET Toolkit* (Rossman 2000). The technique involves the following

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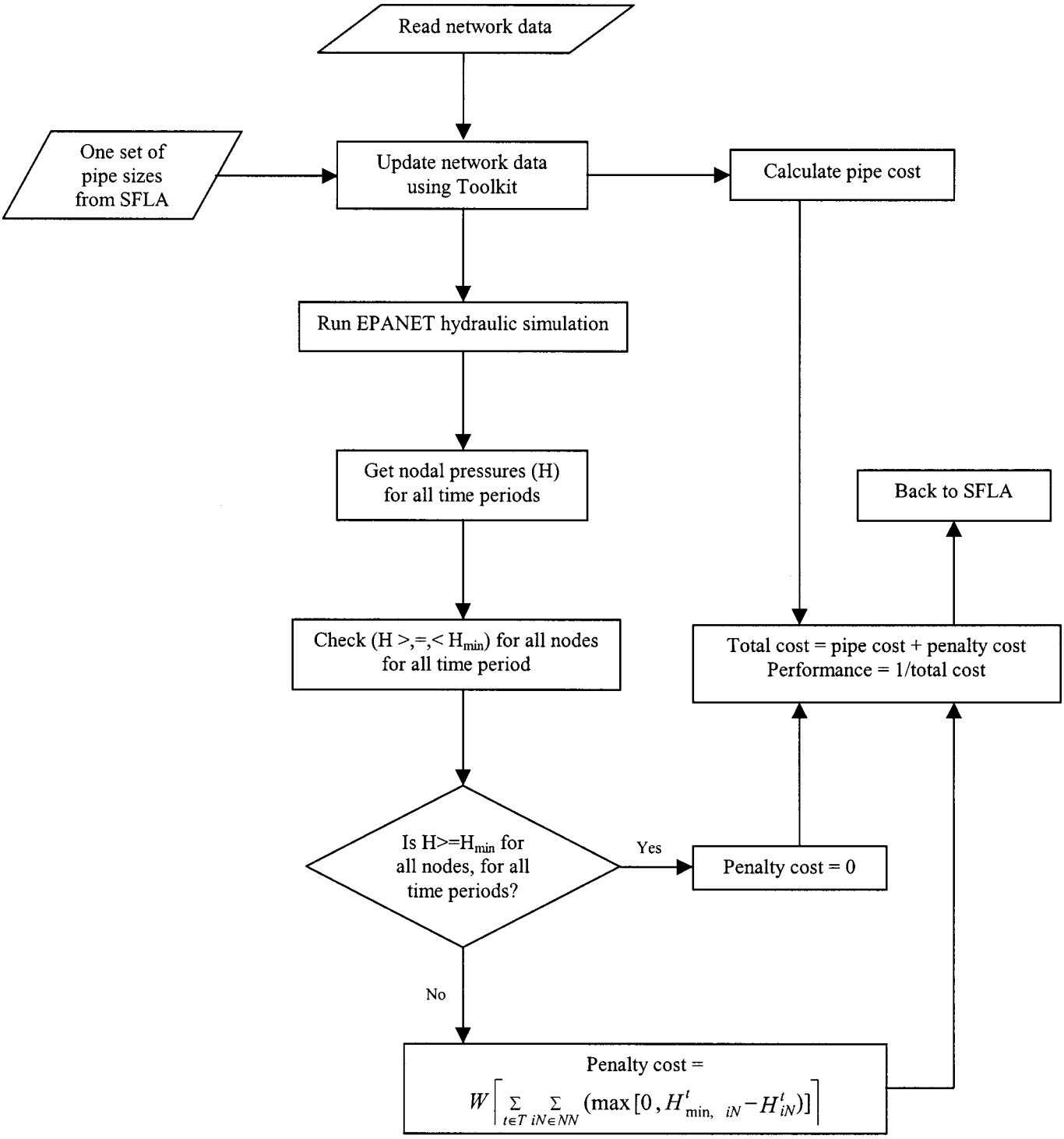


Fig. 2. Flow chart for performance evaluation for meme (decision vector) within *SFLANET*

steps for design/augmentation of water distribution systems. Schematic diagrams of the *SFLANET* algorithm are shown in Figs. 1 and 2.

Encoding–Decoding of Set of Design Solution

Memotype(s), the actual contents of a meme are treated as the decision variables for the network design problem. An *npipe*-dimensional problem (*npipe* pipes) means a frog with a meme comprised of "*npipe*" memotypes. In this application, each memotype of a meme is a numeric representation of a discrete pipe diameter. For example, if we have eight available pipe sizes to select from for a pipe, then the memotypes are encoded as 1–8 where 1 corresponds to the smallest pipe diameter and 8 corresponds to the largest pipe diameter. For a network of four pipes, a frog can have a meme comprised of four memotypes for the four pipes, e.g., (2, 4, 3, and 7). *SFLA* uses the value of the memotypes for the evolution. The memotypes are decoded to actual pipe sizes according to their numeric representation before they are passed to the hydraulic simulation model.

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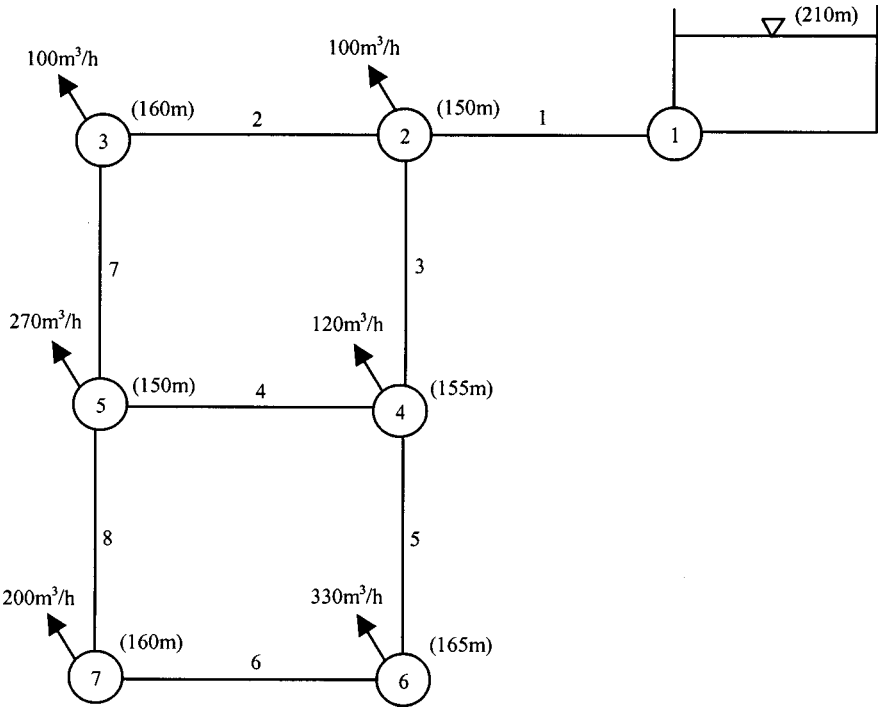


Fig. 3. Alperovits and Shamir network (all pipes are 1,000 m long)

For the description of the following *SFLA* steps, the readers should keep in mind that the algorithm is using an abstract model where a frog's position (meme) is dictated by a selected set of pipe sizes at a particular time and the position change means a change in pipe sizes.

Choice of Initial Set of Design Solutions

To begin the search, a population of F virtual frogs (sets of solutions) is randomly initialized with positions [f(encoded meme of *npipe* memotypes)] within the feasible space Ω. The position of the "ith" *frog* with *npipe* memotypes is represented as $\mathbf{D(i)} = (D_i^1, D_i^2, \ldots, D_i^{npipe})$.

Performance Evaluation of Virtual Frog

The performance of each frog is then computed based on its position. Since *SFLA* is an unconstrained optimization method, performance includes the solution cost and some penalty reflecting constraint violation. The first step of the performance evaluation is that *EPANET* via *Toolkit* solves the conservation of mass and energy equations and computes the nodal pressure heads for each demand.

The minimum pressure constraint separates feasible solutions from infeasible ones. During the search infeasible solutions violating pressure constraints are not ignored. These solutions (frogs) are considered in the process to make the exploration more flexible but do incur a penalty cost reflecting the constraint violation.

A simple additive penalty function is applied, $W^*[\max(0, H_{\min,iN} - H_{iN})]$ where $H_{\min,iN}$ and H_{iN} = minimum allowable pressure and the simulated pressure at node iN, respectively, and W = penalty multiplier per unit violation for the specific problem. The total cost (TC) is then

$$\mathrm{TC}(D_1, \ldots, D_{npipe}) = \sum_{k \in npipe} C(D_k, L_k) + W\left[\sum_{t \in T} \sum_{iN \in NN} [\max(0,\ H_{\min,iN} - H_{iN})]\right] \tag{7}$$

The penalty multiplier, W, is chosen to normalize nominal values of the penalties to the same scale as the pipe cost. The performance function, f, to be maximized in *SFLANET* is 1/TC. The detailed steps for computing performance are described below and shown in Fig. 2.

Table 1. Cost Data for New Pipes for Alperovits and Shamir Network

Memotype value	Diameter (in.)	Diameter (mm)	Cost ($/m)
1	1	25.4	2
2	2	50.8	5
3	3	76.2	8
4	4	101.6	11
5	6	152.4	16
6	8	203.2	23
7	10	254.0	32
8	12	304.8	50
9	14	355.6	60
10	16	406.4	90
11	18	457.2	130
12	20	508.0	170
13	22	558.8	300
14	24	609.6	550

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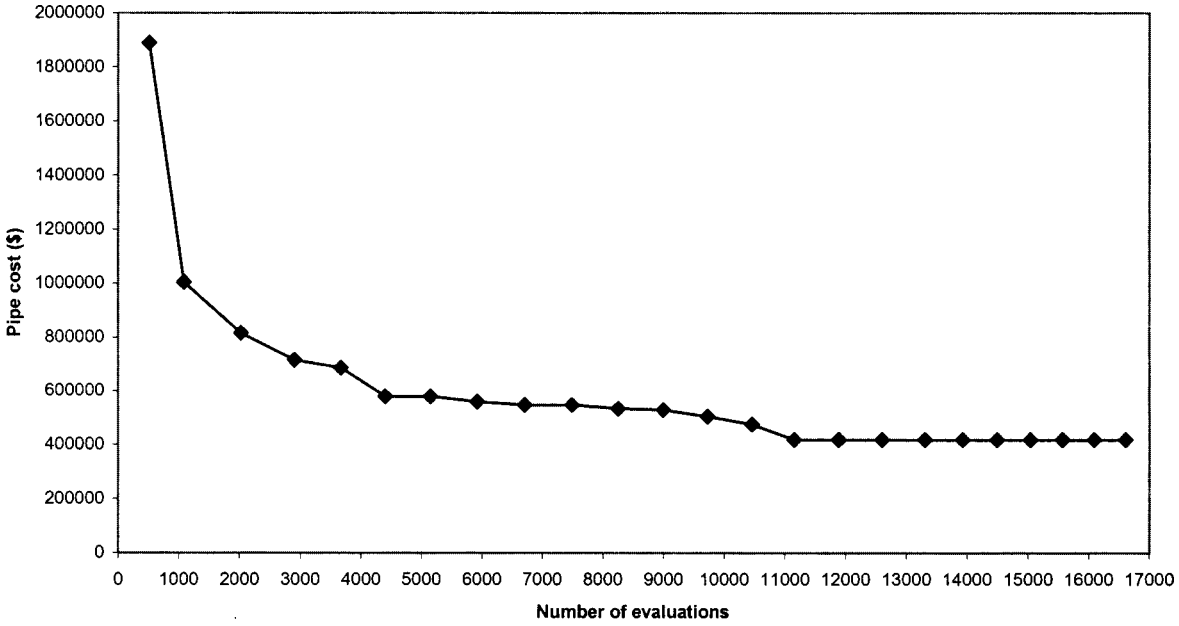


Fig. 4. Evolution of solution cost for Alperovits and Shamir network using *SFLANET*

1. *Step 0*:
   Input network, cost, and performance function data, and required minimum nodal pressure heads.
2. *Step 1*:
   A set of pipe diameters for frog *i*, $\mathbf{D(i)}$, $(\mathbf{D_i^{iP}}; iP = 1,...,npipe)$ (i.e., decoded memetic information containing *npipe* pipe sizes) is passed to the performance evaluation program.
3. *Step 2*:
   Total pipe cost $[f(\text{length,diameter})]$ is calculated using the first part of Eq. (7). The network data is updated for $\mathbf{D(i)}$ using *EPANET Toolkit*.
4. *Step 3*:
   Hydraulic simulation (*EPANET*) is run and the nodal pressures are obtained for each demand pattern.
5. *Step 4*:
   If the solution does not satisfy minimum head constraints, the penalty cost is evaluated as the product of the penalty multiplier (W) and magnitude of the nodal head violation summed for all nodes and all demand patterns.
6. *Step 5*:
   The total cost for frog *i*, TC_i, is computed as the sum of pipe cost and penalty cost [Eq. (7)].

Table 2. Solutions for the Alperovits and Shamir Network

	Alperovits and Shamir (1997)		Goulter et al. (1986)		Kessler and Shamir (1989)		Eiger et al. (1994)		Savic and Walters (1997)		Cunha and Sousa (1999)	*SFLANET*
Pipe	L (m)	D (in.)	L (m)	D (in.)	L (m)	D (in.)	L (m)	D (in.)	D^a (in.)	D^b (in.)	D^c (in.)	D (in.)
1	256.0	20	383.0	20	1,000.0	18	1,000.0	18	18	20	18	18
	744.0	18	617.0	18								
2	996.4	8	1,000.0	10	66.0	12	238.0	12	10	10	10	10
	3.6	6			934.0	10	762.0	10				
3	1,000.0	18	1,000.0	16	1,000.0	16	1,000.0	16	16	16	16	16
4	319.4	8	687.0	6	713.0	3	1,000.0	1	4	1	4	4
	680.6	6	313.0	4	287.0	2						
5	1,000.0	16	1,000.0	16	836.0	16	628.9	16	16	14	16	16
					164.0	14	371.1	14				
6	784.9	12	98.0	12	109.0	12	989.1	10	10	10	10	10
	215.1	10	902.0	10	891.0	10	11.0	8				
7	1,000.0	6	492.0	10	819.0	10	921.9	10	10	10	10	10
			508.0	8	181.0	8	78.1	8				
8	990.9	6	20.0	2	920.0	3	1,000.0	1	1	1	1	1
	9.07	4	980.0	1	80.0	2						
($)[d]	497.5		435.0		417.5		402.4		419.0	420.0	419.0	419.0

[a]Solution obtained using GA and $\omega = 10.5088$.

[b]Solution obtained using GA and $\omega = 10.9031$.

[c]Solution obtained using simulated annealing (SA).

[d]Total pipe cost in thousands.

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Table 3. Nodal Pressures for Optimal Solution of Alperovits and Shamir Network Obtained by *SFLANET*

Node	Pressure[a] (m)
2	53.24
3	30.44
4	43.44
5	33.78
6	30.43
7	30.53

[a]Pressure must be equal or greater than 30.00 m.

7. *Step 6*:
   Performance of the set of decision variables is computed as $f(i)=1/\mathrm{TC}_i$. This value is passed to *SFLA*.

This process is completed for all possible sets of solutions (frogs).

Rank and Partition the Frogs into Memeplexes

After the performance for all frogs is calculated, the frogs are sorted in order of decreasing performance value. The sorted frogs are stored in an array $\mathbf{X}=\{\mathbf{D(ir)},f(ir),ir=1,...,F\}$, where $ir=1$ represents the frog with the best performance value (i.e., least-cost solution set). The best frog's position in the entire population of F frogs, $\mathbf{D_X}$ is recorded. X is then partitioned into m memeplexes $\mathbf{Y^1},\mathbf{Y^2},\ldots,\mathbf{Y^m}$, each containing n frogs, such that

$$\mathbf{Y^k}=[\mathbf{D(j)^k},f(j)^k|\mathbf{D(j)^k}=\mathbf{D[k+m(j-1)]}$$
$$f(j)^k=f[k+m(j-1)],\quad j=1,...,n,\quad k=1,...,m \tag{8}$$

For example, for $m=3$, frog ranking 1 goes to memeplex 1, frog ranking 2 goes to memeplex 2, frog ranking 3 goes to memeplex 3, frog ranking 4 goes to memeplex 1, and so on. A memeplex is analogous to a complex in *SCE*.

Memetic Evolution within Memeplex by the Frog Leaping Algorithm (FLA)

As described earlier each memeplex acts and evolves as an independent culture. Details of the frog leaping algorithm within each memeplex that improves the frogs' performance towards the optimum are presented below.

1. *Step 0*:
   Set $im=0$ where im counts the number of memeplexes and will be compared to the total number of memeplexes m. Set $iN=0$ where iN counts the number of evolutionary steps

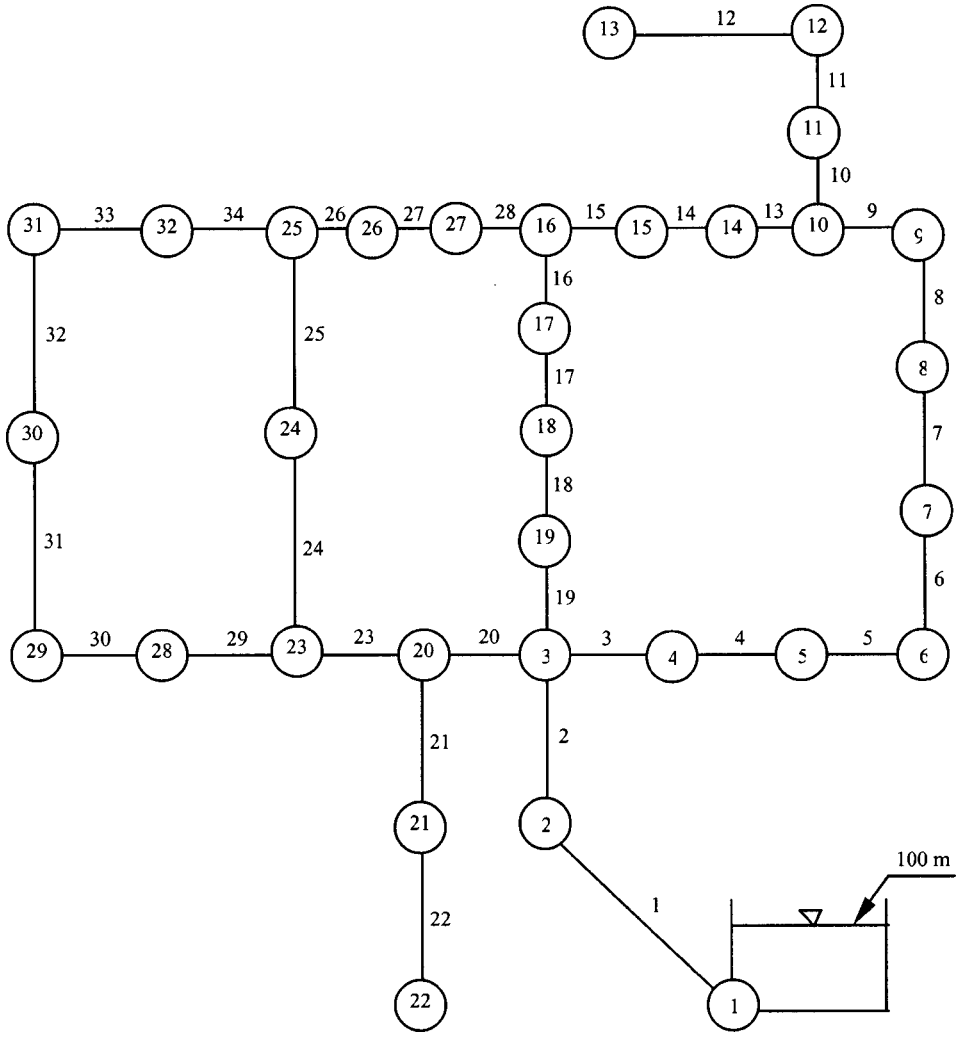


Fig. 5. Hanoi network (Fujiwara and Khang 1990)

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Table 4. Data for Hanoi Network

Node Data		Pipe Data	
Node	Demand (m^3/h)	Pipe	Length (m)
1 (source)	−19,940	1	100
2	890	2	1,350
3	850	3	900
4	130	4	1,150
5	725	5	1,450
6	1,005	6	450
7	1,350	7	850
8	550	8	850
9	525	9	800
10	525	10	950
11	500	11	1,200
12	560	12	3,500
13	940	13	800
14	615	14	500
15	280	15	550
16	310	16	2,730
17	865	17	1,750
18	1,345	18	800
19	60	19	400
20	1,275	20	2,200
21	930	21	1,500
22	485	22	500
23	1,045	23	2,650
24	820	24	1,230
25	170	25	1,300
26	900	26	850
27	370	27	300
28	290	28	750
29	360	29	1,500
30	360	30	2,000
31	105	31	1,600
32	805	32	150
		33	860
		34	950

Table 5. Cost Data for Hanoi Network

Memotype value	Diameter (in.)	Diameter (mm)	Cost ($/m)
1	12	304.8	45.726
2	16	406.4	70.400
3	20	508.0	98.378
4	24	609.6	129.333
5	30	762.0	180.748
6	40	1,016.0	278.280

and will be compared to maximum number of steps, N, to be completed within each memeplex. N and m are defined by the modeler.

2. *Step 1*:
   Set $im = im + 1$.
3. *Step 2*:
   Set $iN = iN + 1$.
4. *Step 3*:
   Construct a submemeplex: A frogs' goal is to move towards the optimum location by improving its memes. As stated earlier, frogs can adapt their ideas from the best frog within the memeplex or from the population's global best. Regarding the selection of the memeplex best, it is not always desirable to use the best frog because the frogs' tendency would be to concentrate around that particular frog that might be a local optimum. So, a subset of the memeplex called a submemeplex is considered. However, the submemeplex selection strategy gives higher weights to the frogs that show more promise by having higher performance values and less weight to those with lower performance values. The weights are assigned with a triangular probability distribution, i.e.

$$p_j = 2(n+1-j)/[n(n+1)], \quad j=1,...,n$$

such that, within a memeplex the frog with the best performance has the highest probability of being selected for the submemeplex, $p_1 = 2/(n+1)$ and the frog with the worst performance has the lowest probability, $p_n = 2/[n(n+1)]$. q distinct frogs are then randomly selected from n frogs in the memeplex to form the submemeplex array **Z**. The submemeplex is sorted so that frogs are arranged in order of decreasing performance value. Record the best $(iq=1; iq=1,...,q)$ and worst $(iq=q)$ frog's position in the submemeplex as the $\mathbf{DS_B}$ and $\mathbf{DS_W}$ vectors, respectively.

5. *Step 4*:
   Improve the worst frog's position: The step size (as a vector for all the memotypes) and new position for the frog with the worst performance in the submemeplex are computed by

$$\text{Step size, } \mathbf{d} = MIN\{INT[rand*(\mathbf{DS_B}-\mathbf{DS_W})], \mathbf{d_{max}}\} \quad \text{for positive step}$$

$$= MAX\{INT[rand*(\mathbf{DS_B}-\mathbf{DS_W})], -\mathbf{d_{max}}\} \quad \text{for negative step}$$

   where $rand$ = random number in the range [0,1] and $\mathbf{d_{max}}$ = maximum step size allowed (maximum incremental change in pipe diameter value) for a frog after being infected. The new position is then computed by

$$\mathbf{D_{(iq=q)}} = \mathbf{D_W} + \mathbf{d} \tag{9}$$

   If $\mathbf{D_{(iq=q)}}$ is within the feasible space, Ω, compute the new performance value $f_{(iq=q)}$. Otherwise go to *step 5*.
   If the new $f_{(iq=q)}$ is better than the old $f_{(iq=q)}$, i.e., if the evolution produces a better solution, then replace the old $\mathbf{D_{(iq=q)}}$ with the new one and go to *step 7*. Otherwise go to *step 5*.
6. *Step 5*:
   If *step 4* cannot produce a better solution then the step and new position are computed for that frog based upon the present global optimal solution by

$$\text{Step size, } \mathbf{d} = MIN\{INT[rand*(\mathbf{D_X}-\mathbf{DS_W})], \mathbf{d_{max}}\} \quad \text{for positive step}$$

$$= MAX\{INT[rand*(\mathbf{D_X}-\mathbf{DS_W})], -\mathbf{d_{max}}\} \quad \text{for negative step}$$

   and the new position is computed by Eq. (9).
   If $\mathbf{D_{(iq=q)}}$ is within the feasible space, Ω, compute the new performance value $f_{(iq=q)}$ otherwise go to *step 6*.
   If the new $f_{(iq=q)}$ is better than the old $f_{(iq=q)}$, i.e., if the evolution produces a better solution, then replace the old $\mathbf{D_{(iq=q)}}$ with a new one and go to *step 7*. Otherwise go to *step 6*.
7. *Step 6*:
   Censorship: If the frog's new position is either infeasible or not better than the old position using both criteria, the spread of defective meme is stopped by randomly generating a new frog '*r*' at a feasible location to replace the frog

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Table 6. Solutions for Hanoi Network Problem Obtained by Using Discrete Diameter Techniques

	Savic and Walters (1997) (GA with $\omega = 10.5088$)		Savic and Walters (1997) (GA with $\omega = 10.9031$)		Cunha and Sousa (1999) (Simulated Annealing)	
Pipe/node number	Diameter (mm)	Pressure (m)	Diameter (mm)	Pressure (m)	Diameter (mm)	Pressure (m)
1	1,016	100.00	1,016	100.00	1,016	100.00
2	1,016	97.14	1,016	97.14	1,016	97.14
3	1,016	61.63	1,016	61.63	1,016	61.63
4	1,016	56.83	1,016	57.26	1,016	56.82
5	1,016	50.89	1,016	51.86	1,016	50.86
6	1,016	44.62	1,016	46.21	1,016	44.57
7	1,016	43.14	1,016	44.91	1,016	43.10
8	1,016	41.38	1,016	43.40	1,016	41.33
9	1,016	39.97	762	42.23	1,016	39.91
10	762	38.93	762	38.79	762	38.86
11	609.6	37.37	762	37.23	609.6	37.30
12	609.6	33.94	609.6	36.07	609.6	33.87
13	508	29.72[a]	406.4	31.86	508	29.66[a]
14	406.4	35.06	406.4	33.19	406.4	34.94
15	304.8	33.07	304.8	32.90	304.8	32.88
16	304.8	30.15	406.4	33.01	304.8	29.79[a]
17	406.4	30.24	508	40.73	406.4	29.95[a]
18	508	43.91	609.6	51.13	508	43.81
19	508	55.53	609.6	58.03	508	55.49
20	1,016	50.39	1,016	50.63	1,016	50.43
21	508	41.03	508	41.28	508	41.07
22	304.8	35.86	304.8	36.11	304.8	35.90
23	1,016	44.15	1,016	44.61	1,016	44.24
24	762	38.84	762	39.54	762	38.50
25	762	35.48	762	36.40	762	34.79
26	508	31.46	508	32.93	508	30.87
27	304.8	30.03	304.8	32.18	304.8	29.59[a]
28	304.8	35.43	304.8	36.02	304.8	38.60
29	406.4	30.67	406.4	31.38	406.4	29.64[a]
30	406.4	29.65[a]	406.4	30.47	304.8	29.90[a]
31	304.8	30.12	304.8	30.95	304.8	30.18
32	304.8	31.36	304.8	32.24	406.4	32.64
33	406.4		406.4		406.4	
34	508		508		609.6	
Cost[b]		6.073		6.195		6.056

[a]Infeasible solutions (pressure $<$ 30 m) when solved by *EPANET*.

[b]Total pipe cost in $ million.

whose new position was not favorable to progress towards an optimum value. Compute $f(r)$ and set $\mathbf{D}_{(\mathbf{iq}=\mathbf{q})} = r$ and $f_{(iq=q)} = f(r)$.

8. *Step 7*:
   Upgrade the memeplex: After the memetic change for the worst frog in the submemeplex, replace $\mathbf{Z}$ in their original locations in $\mathbf{Y^k}$. Sort $\mathbf{Y^k}$ in order of decreasing performance value.
9. *Step 8*:
   If $iN < N$, go to *step 2*.
10. *Step 9*:
    If $im < m$, go to *step 1*. Otherwise go to shuffling operation to form new memeplex sets.

Shuffling of Population

After N evolutions within each memeplex the population is shuffled. Periodic shuffling strategy promotes a global exchange of information among the frogs and helps to concentrate the search in the direction of the most promising region identified by individual memeplexes.

Here the newly evolved $\mathbf{Y^1},\ldots,\mathbf{Y^m}$ are replaced in $\mathbf{X}$, such that $\mathbf{X} = \{\mathbf{Y^k}, k = 1,...,m\}$. $\mathbf{X}$ is then sorted in order of decreasing performance value and partitioned into m memeplexes using Eq. (8). The best frog's position in the entire population (F frogs), $\mathbf{D_X}$, is updated. Memetic evolution within the memeplexes is repeated until a stopping criteria is met.

Stopping Criteria

Two stopping criteria are used in *SFLA*. The first limits the number of evaluations or shuffle operations. The second, and more desirable result, is the number of consecutive shuffle operations over which the global best solution has not changed which implies convergence to an optimal solution.

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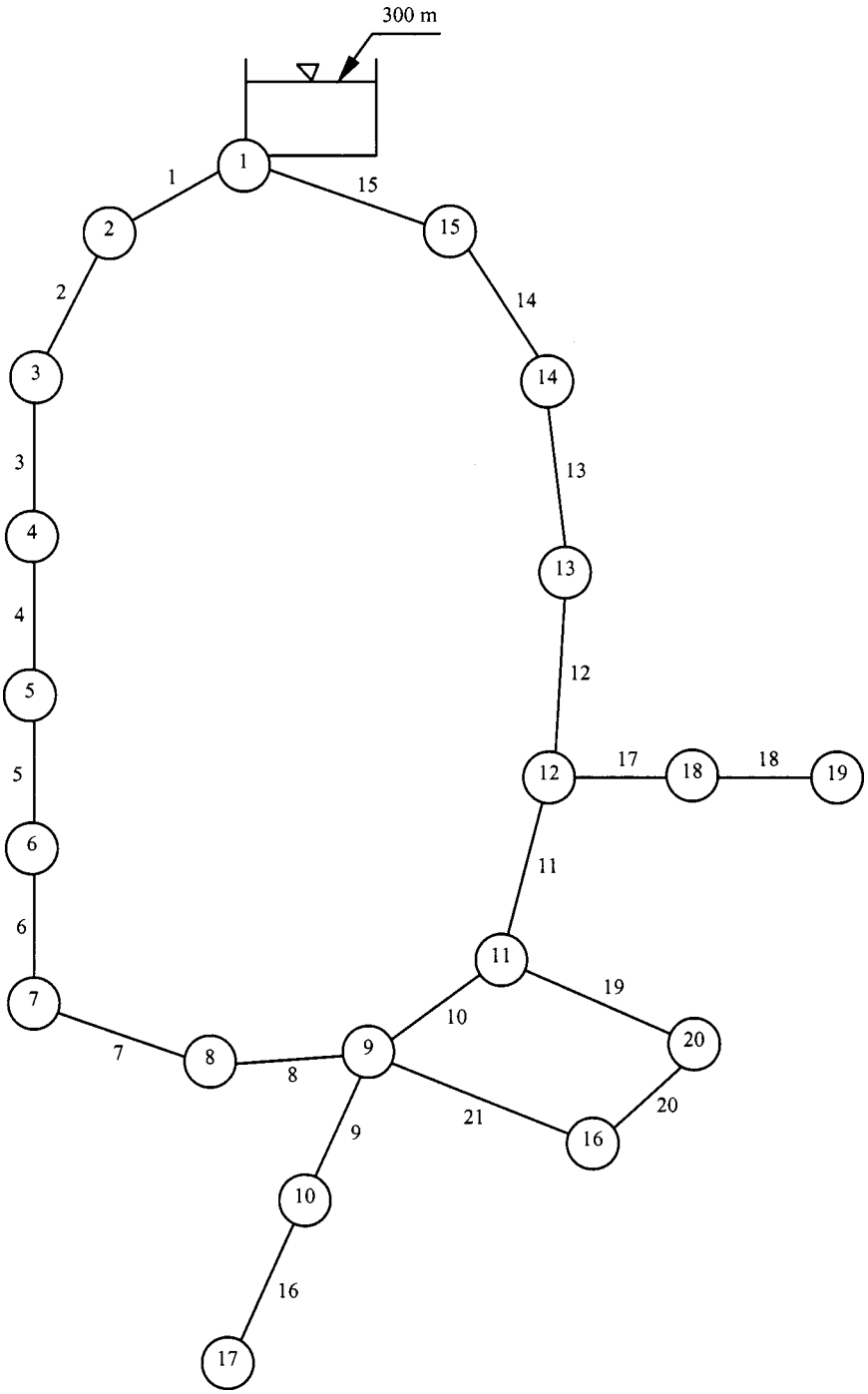


Fig. 6. New York City water supply tunnel system (Schaake and Lai 1969)

Selection of Shuffled Frog Leaping Algorithm Parameters

Like other heuristics the performance of *SFLA* is sensitive to the parameters used. The primary *SFLA* parameters are the number of memeplexes (m), the number of frogs per memeplex (n), the number of frogs per submemeplex (q), the number of memetic evolutions within a memeplex before shuffling (N), and the maximum allowable step size ($d_{\max}$). A sensitivity study on the parameters has been completed by the authors (Eusuff and Lansey 2000). Since no general guidelines to choose these parameters have been identified, a parameter sensitivity analysis was completed to determine a proper set of parameters for each test network. After the good parameters were identified, convergence tests were performed by varying the *SFLA* parameters for the different networks. Results for the New York City Water Supply Tunnel System are presented in the application section.

SFLANET Applications

Alperovits and Shamir Network

In order to demonstrate the efficacy of *SFLANET*, three networks that have appeared in the literature were designed. The first system, a two-loop network, was introduced by Alperovits and Shamir (1977). It was also examined by Quindry et al. (1981);

Table 7. Network Data for New York City Tunnel System

Node Data			Pipe Data		
Node	Demand (ft^3/s)	Minimum head (ft)	Pipe	Length (ft)	Diameter[a] (ft)
1 (source)	−2,017.5	300.0	1	11,600	180
2	92.4	255.0	2	19,800	180
3	92.4	255.0	3	7,300	180
4	88.2	255.0	4	8,300	180
5	88.2	255.0	5	8,600	180
6	88.2	255.0	6	19,100	180
7	88.2	255.0	7	9,600	132
8	88.2	255.0	8	12,500	132
9	170.0	255.0	9	9,600	180
10	1.0	255.0	10	11,200	204
11	170.0	255.0	11	14,500	204
12	117.1	255.0	12	12,200	204
13	117.1	255.0	13	24,100	204
14	92.4	255.0	14	21,100	204
15	92.4	255.0	15	15,500	204
16	170.0	260.0	16	26,400	72
17	57.5	272.8	17	31,200	72
18	117.1	255.0	18	24,000	60
19	117.1	255.0	19	14,400	60
20	170.0	255.0	20	38,400	60
			21	26,400	72

[a]Existing pipe diameters.

Fujiwara et al. (1987); Kessler and Shamir (1989, 1991); Bhave and Sonak (1992); Savic and Walters (1997); and Cunha and Sousa (1999). The system layout and network data are presented in Fig. 3. The minimum allowable pressure is 30 m above ground level for all nodes. Table 1 lists the pipe cost per meter for the 14 available pipe diameters.

After some testing, the *SFLA* parameters were defined as $m=30$, $n=17$, $q=9$, $N=17$, and $d_{\max}=100\%$ of the available pipe diameter range. Each frog is characterized by a meme of 8 memotypes where each memotype can assume any of the encoded values of the 14 available pipe sizes. Thus 14^8 types of ideas are possible among the virtual frogs.

Fig. 4 shows the evolution towards the optimal solution in *SFLANET* for one of several runs. The best solution found using *SFLANET* is the same as Savic and Walters (1997) and Cunha and Sousa (1999) (Table 2). The optimum cost \$419,000 was found using five different random initial points. However, *SFLANET* found the optimal solution more quickly than previous techniques. The least number of function evaluations was 11,155 (with 11,323 and 11,692 runs being the average and the most number of evaluations required, respectively) using *SFLANET* compared to around 65,000 evaluations for the genetic algorithm (GA) (Savic and Walters 1997) and around 25,000 evaluations for simulated annealing (SA) (Cunha and Sousa 1999). For reference, four continuous diameter solutions are also shown in Table 2. Since a discrete solution is not required these solutions are lower than the *SFLANET* cost.

Savic and Walters (1997) presented two discrete diameter solutions obtained by using GA. These two solutions are the optimal designs found for ω [see Eq. (4)] equal to 10.5088 and 10.9031, respectively. The higher ω results in higher head losses requiring larger pipes in the final design. This issue is more critical in larger systems. Table 3 lists the nodal pressures for the optimum design found by using *SFLANET*. The ω value in *EPANET* v. 2.0 is 10.667.

Table 8. Cost Data for New Pipes for New York City Tunnel System Problem

Memotype value	Diameter (in.)	Cost (\$/ft)
1	48	134
2	60	176
3	72	221
4	84	268
5	96	316
6	108	365
7	120	416
8	132	469
9	144	522
10	156	577
11	168	632
12	180	689

Hanoi Network

The Hanoi network shown in Fig. 5 was first presented by Fujiwara and Khang (1990). Its system data are summarized in Table 4. The minimum allowable nodal pressure is 30 m above ground level for all nodes. Table 5 lists the pipe cost per meter for six available pipe sizes. With these six discrete pipe sizes for a 34 pipe network design, the total solution space consists of 6^{34} different possible network designs. Results are presented using the following *SFLA* parameters: $m=30$, $n=30$, $q=15$, $N=15$, and $d_{\max}=100\%$ of the available pipe diameter range.

The network was solved using *EPANET Toolkit* for the three previously published discrete diameter solutions listed in Table 6. The resulting nodal pressures are given in columns (3), (5), and (7) in Table 6. Only one solution (GA with $\omega=10.9031$) is feasible.

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Table 9. Solutions for New York City Tunnel Expansion Problem Obtained by Using Discrete Diameter Methods

Pipe	Gessler (1985) *D* (in.)	Morgan and Goulter (1985) *D* (in.)	Murphy et al. (1993) *D* (in.)	Savic and Walters (1997)[a] *D* (in.)	Savic and Walters (1997)[b] *D* (in.)	Lippai et al. (1999) and *SFLANET* *D* (in.)
1	—	—	—	—	—	—
2	—	—	—	—	—	—
3	—	—	—	—	—	—
4	—	—	—	—	—	—
5	—	—	—	—	—	—
6	—	—	—	—	—	—
7	100	144	—	108	—	132
8	100	—	—	—	—	—
9	—	—	—	—	—	—
10	—	—	—	—	—	—
11	—	—	—	—	—	—
12	—	—	—	—	—	—
13	—	—	—	—	—	—
14	—	—	—	—	—	—
15	—	—	120	—	144	—
16	100	96	84	96	84	96
17	100	96	96	96	96	96
18	80	84	84	84	84	84
19	60	60	72	72	72	72
20	—	—	—	—	—	—
21	80	84	72	72	72	72
Cost[c]	41.80	39.20	38.80	37.13	40.42	38.13

[a]Solution obtained by using genetic algorithm and $\omega = 10.5088$. This solution results in infeasible pressures at nodes 16, 17, and 19 when the network was simulated using $\omega = 10.667$ which was used for *SFLANET*.

[b]Solution obtained by using genetic algorithm and $\omega = 10.9031$.

[c]Total pipe cost in $ million.

SFLANET was then used to find the optimum network design with allowable minimum pressure 29.59 m [minimum among the violated nodes in Table 6 for the Cunha and Sousa (1999) solution] and 30.00 m. The solutions obtained by using *SFLANET* are equivalent to results presented by Savic and Walters (1997) in Table 6 (equivalent to column 2 for $H_{min} = 29.59$ m and equivalent to column 4 for $H_{min} = 30.00$ m). Although the best solution found for five runs from different random starting points using *SFLANET* is slightly more expensive than Cunha and Sousa (1999), *SFLANET* was more efficient. The optimum cost of $6,073,000 was found in a minimum of 26,987 evaluations (with 27,546 and 28,309 runs being the average and the most number of evaluations required, respectively) using *SFLANET* compared to 53,000 evaluations required by simulated annealing. Three small pipes (pipe No. 30, 32, and 34) were different by one diameter increment. Two of those pipes had diameters near the boundary. This condition suggests that *SFLA* may have problems in these cases. Further research is recommended for the problems when the decisions are near their limits.

Parallel Expansion of New York City Water Supply Tunnel

The New York City water supply tunnel system, shown in Fig. 6 was first described in detail and optimized by Schaake and Lai (1969). Unlike the previous examples, this system is in place and requires expansion. A number of other researchers have examined this problem (Quindry et al. 1981; Morgan and Goulter 1985; Bhave 1985; Murphy et al. 1993; Savic and Walters 1997; Lippai et al. 1999). The imperial system of units was used here to enable comparisons with previous studies. Existing pipe and node data are listed in Table 7. Due to pipe aging, the existing gravity flow tunnels are inadequate to meet the pressure requirements (nodes 16, 17, 18, 19, and 20) for the projected demands. Therefore new pipes can be added in parallel to the existing pipes to meet the minimum pressure head requirements.

For this problem 16 possible decisions are available including 15 pipe sizes and the ''do nothing'' option. Considering all 21 pipes for possible duplication results in 16^{21} possible designs. Table 8 lists the unit pipe costs. The *C* value for new pipes was taken as 100. *SFLA* parameters were $m = 30$, $n = 30$, $q = 22$, $N = 22$, and $d_{max} = 100\%$ of the available pipe diameter range. Results of varying these parameters is described below.

The best discrete solutions found in previous studies and that obtained by using *SFLANET* are summarized in Table 9. The figures listed in the table are the new pipe diameters to be added in parallel to the respective existing lines. The solution was found by *SFLANET* performing several runs using different random starting points using the parameters noted above. The optimal solution found by *SFLANET* is $38.13 million, and was determined in 31,267 evaluations (the least among five runs with 100% success). In terms of efficiency *SFLA* is faster than the GA used by Lippai et al. (1999) that determined its best solution ($38.13 million) after 46,016 evaluations. This solution allowed for a violation of 0.05 ft so the minimum head is 0.02 ft below the requirement as determined by *EPANET* (v. 2.0).

Table 10 lists the optimal nodal head values for the *SFLANET* solution. It is important to note that the solutions found by Savic and Walters (1997) ($37.13 million with $\omega = 10.5088$) resulted in

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Table 10. Nodal Pressures for Optimal Solution of New York City Tunnel System Obtained by *SFLANET*

Node	Pressure[a] (ft)
1 (source)	300.00
2	294.23
3	286.20
4	283.84
5	281.76
6	280.15
7	277.61
8	276.56
9	273.70
10	273.66
11	273.79
12	275.07
13	278.04
14	285.53
15	293.31
16	260.00 (259.80)[b]
17	272.80 (272.60)
18	261.11
19	254.98 (254.80)
20	260.65

[a]Minimum nodal pressure requirement for all nodes except 16 and 17 is 255 ft, for nodes 16 and 17 it is 260 and 272.8 ft, respectively.

[b]Values in parentheses for nodes 16, 17, and 19 are pressures when the network was simulated using solutions obtained by Savic and Walters (1997) ($\omega = 10.5088$) and with $\omega = 10.667$ which was used for *SFLANET*.

infeasible pressures at nodes 16, 17, and 19 when the network was simulated using the $\omega = 10.667$ that is used for the *EPANET* (v. 2.0). To satisfy minimum pressure requirements without minor violations, *SFLANET* found the solution by Murphy et al. (1993) ($38.8 million) in eight of eight runs with different initial seeds requiring between 21,569 and 24,817 function evaluations.

The *SFLA* parameters used in the above problems were selected after some testing. Like all heuristics, the best parameter values are problem dependent. The effect of the parameter selection on convergence to the global optimal and on the number of evaluations was examined for the NY tunnel problem (Table 11). Four parameters were studied: the number of memeplexes, n; the number of frogs per memeplex, m; the number of evolutions between shuffles, N; and the maximum stepsize, $d_{\max}$. In terms of finding the global optimum, the method was insensitive to N and $d_{\max}$. But the number of function evaluations decreased as $d_{\max}$ was increased since decreasing the step size limited movement toward the solution. *SFLA* was more sensitive to n and m. It appears that increasing either parameter increases the likelihood of finding the global solution but at a cost of more function evaluations. All of these runs allowed the slight pressure violation (0.05 ft). Since the solution has pressure heads very near the bound value (nodes 16, 17, and 19), convergence to the apparent global optimum is slower than the Murphy solution.

Conclusions

The computational complexity in solving the optimal network design problem is extremely high. Water distribution network design belongs to a group of inherently intractable problems referred to as NP-hard problems. Traditional optimization methods have been limited by their ability to handle the complex nature of the problem and have only been applied with simplifications and assumptions.

Table 11. Results of *SFLANET* Parameter Selection Test for New York City Tunnel System

Test number	Parameter name	Parameter value	Average number of evaluations required to achieve global optimum[a]	Range of evaluations to achieve global optimum	Success rate (%) in five runs
Test 1	Frogs/memeplex, n	10	Not found	Not found	0
		20	26,000	25,468–26,522	40
		30	31,400	31,267–31,713	100
		40	38,900	37,437–40,894	100
Test 2	Number of memeplex, m	10	Not found	Not found	0
		20	27,500	26,813–28,983	60
		30	31,400	31,267–31,713	100
		40	38,300	36,841–40,263	100
Test 3	Number of evolutions in a memeplex between two successive shuffling, N	10	37,900	36,437–39,836	80
		15	32,400	32,018–33,145	100
		22	31,400	31,267–31,713	100
		30	41,100	39,542–43,211	100
Test 4	Maximum step size as some percent of pipe size range, $d_{\max}$	25	61,200	59,058–64,015	100
		50	45,300	43,533–47,619	100
		75	40,100	38,572–42,156	100
		100	31,400	31,267–31,713	100

Note: Base values are $n = 30$, $m = 30$, $N = 22$, and $d_{\max} = 100\%$ of the pipe size range.

[a]Rounded to the nearest hundred (00).

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Techniques have been developed in the field of stochastic optimization such as genetic algorithm and simulated annealing that have been applied to the optimal network design problem and retain the reality of the physical system. This paper described the development of *SFLANET*, an optimization model based on *shuffled frog leaping algorithm* (*SFLA*), a recently developed meta-heuristic for discrete problems. It is a population based, cooperative search metaphor inspired by natural memetics. *SFLANET* can be used to assist engineers develop designs for water distribution networks or expansions of existing networks. To model the network hydraulics, *EPANET* is directly linked to *SFLA*.

In this paper, the efficacy of *SFLANET* was tested by solving three water distribution network optimization problems. *SFLANET* found previous best solutions for the two example networks and a very nearly optimal solution for the third. In solving these problems *SFLA* found the optimal solutions in fewer iterations than genetic algorithms and simulated annealing. The robustness of the method was also examined with respect to *SFLA* parameters.

The *SFLANET* framework is very versatile and can be extended to design pumps or other network components. Also, although *SFLANET* uses *EPANET* as the network hydraulic solver, any available hydraulic model can be incorporated for that purpose.

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